

Ashraful Islam

AI/ML Engineer Aspirant | Python Backend Developer | RAG & Generative AI
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🌐 ashraful-islam | 🌐 github.com/ashraf1600 | 🌐 Portfolio

PROFESSIONAL SUMMARY

CSE undergraduate at **CUET** and aspiring **AI Engineer** with research experience in **RAG**, **Explainable ML**, and **Computer Vision**. Remote **ML & DataOps Intern** at **Poridhi.io**, gaining hands-on experience in **AWS networking** and **MLOps**. Skilled in **Python**, machine learning, and backend development.

WORK EXPERIENCE

Poridhi.io

Machine Learning & DataOps Intern

Dhaka, Bangladesh (Remote)

August 2026 – Present

- Gaining hands-on experience with **AWS networking**, **Docker**, and **container networking** while contributing to the setup and troubleshooting of containerized machine learning environments.
- Gaining hands-on experience with **MLflow** for experiment tracking and **Feast** for feature management, contributing to reproducible **MLOps** workflows.
- Exploring practical **machine learning challenges** and contributing to team discussions and problem-solving across model development and deployment workflows.

Synesis IT PLC

Industrial Attachment

Dhaka, Bangladesh

August 2026 – Present

- Building an **AI-powered system** for detecting counterfeit **Bangladeshi banknotes**.

EDUCATION

Chittagong University of Engineering and Technology (CUET)

B.Sc. in Computer Science and Engineering – CGPA: 3.52 / 4.00 (122 Credits Completed)

Chittagong, Bangladesh

2023 – Present

- Undergraduate Thesis:** BondhuAI-BN: Culturally-Aware Bengali Mental Health Support via Scoped Multi-Agent RAG with Domain-Adaptive Fine-Tuning.

Dhaka City College

Higher Secondary Certificate (HSC) – Science – GPA: 5.00 / 5.00

Dhaka, Bangladesh

2019 – 2021

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, JavaScript, SQL, C#
- Generative AI & LLMs:** Agentic AI, LangChain, LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, LLM Fine-Tuning (LoRA), Vector DBs (FAISS, Chroma), Hugging Face, Gemini API
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, Computer Vision (Vision Transformers/DeiT), NLP, Reinforcement Learning (DQN), Explainable AI (SHAP)
- MLOps:** Docker, FastAPI, MLflow, CI/CD Pipelines, Linux, Git/GitHub
- Backend & Databases:** Django, Django REST Framework, PostgreSQL, MySQL, SQLite, MongoDB, RESTful APIs, React

COMPETITIVE PROGRAMMING

- Solved **500+ problems** across platforms – Codeforces (*The_phoenix*, Max Rating: 1133, 335+ solved), LeetCode (*noobashraf21*, 120+ solved), and HackerRank.

FEATURED AI / ML & ENGINEERING PROJECTS

Personal AI Assistant: Agentic Workflows & Automation

GitHub

Tech: Gemini API, LangChain, Streamlit, OOP Architecture, Vector Search, Tool Calling

- Built an interactive AI assistant featuring multi-turn conversational memory, external tool integration, and document search leveraging Gemini API and LangChain.
- Designed with extensible OOP patterns, enabling automated task execution, query routing, and real-time streaming responses in a clean Streamlit interface.

Intelligent Traffic Light Control via Reinforcement Learning

GitHub

Tech: Python, Deep Q-Networks (DQN), SUMO Simulator, PyTorch/TensorFlow, NumPy

- Formulated an adaptive traffic signal control system using Deep Q-Networks (DQN) integrated with the SUMO simulator to dynamically adjust phase duration based on queue density.
- Reduced average vehicle wait times and urban intersection congestion by over 25% compared to static signal timing baselines.

E-Commerce Fraud Detection Platform



Tech: Python, Scikit-Learn, XGBoost, SMOTE-Tomek, Optuna, Flask, Docker

- Developed an end-to-end fraud classification engine handling extreme tabular class imbalance via SMOTE-Tomek resampling and Optuna hyperparameter optimization.
- Packaged model pipelines into a containerized Flask REST API delivering sub-100ms real-time transaction scoring.

ResQNet: Real-Time Disaster Resource Coordination Platform



Tech: Django REST Framework, React, PostgreSQL, Docker, Concurrency Control

- Engineered a disaster resource coordination platform with an auditable need-lifecycle (pending → open → committed → resolved) and atomic conditional updates to prevent race conditions.
- Enforced server-side role-based access control (RBAC) and packaged services with Docker Compose for production deployment.

RESEARCH PUBLICATIONS

- **CCMK-RAG: Content-Conditioned Multi-Key Watermarking for Passage-Level Attribution in Retrieval-Augmented Generation**
Towhidul Islam, Ashraful Islam, Mahfuzul Hoq Chowdhury – *IEEE OMLET 2026* (Accepted).
- **Section-Aware Retrieval-Augmented Generation for Bangladesh Labour Law Question Answering: A Comparative Study**
Towhidul Islam, Ashraf-ul-Islam, Mahfuzulhoq Chowdhury – *IEEE OMLET 2026* (Accepted).
- **An Explainable Machine Learning and Ensemble Voting Classifier for Cyber Attack Detection in Highly Imbalanced IoT Networks**
Ashraful Islam, Towhidul Islam, Dr. Mahfuzulhoq Chowdhury – *IEEE 2026 3rd International Conference on Data Science and Business Systems (ICDSBS 2026)* (Accepted).
- **Vision Transformer-Based Detection of Counterfeit Bangladeshi Banknotes Using the DeiT-Tiny Architecture**
Ashraful Islam, Dr. Mahfuzulhoq Chowdhury, Towhidul Islam – *International Conference on Circuit Power & Computing Technologies (ICCPCT-2026)* (Accepted).
- **A Double Layer Stacking Classifier Based Explainable Tax Evasion Fraud Prediction with Optuna and SMOTE-Tomek**
Dr. Mahfuzulhoq Chowdhury, Saklain Ahmed, Ashraful Islam – *1st International Conference on Next-Generation Electrical & Electronics, Computer Systems, and Technologies (iCONNECT 2026)* (Submitted).
- **A Traffic Rules Violation Prediction Scheme Based on Hybrid Stacking Ensemble Classifier and SHAP Analysis**
Dr. Mahfuzul Hoq Chowdhury, Oarisa Rebayet, Ifath Jeba Chowdhury, Ashraf-ul-Islam, Isnat Mehrin Sami, Snaholata Mondal – *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)* (Copyright Confirmed).

CERTIFICATIONS & TRAINING

- **Full Stack Data Science with Generative AI Bootcamp 1.0** – Inception BD
- **Complete Data Science, Machine Learning, DL, NLP Bootcamp** – Udemy (October 2025)
- **SQL (Intermediate) & SQL (Basic)** – HackerRank (December 2024, Valid to 2034)
- **Database Structures and Management with MySQL** – Meta Professional Certificate
- **Exploratory Data Analysis in Python** – Udemy (February 2026)

LEADERSHIP & EXTRACURRICULARS

- **General Secretary** – Cox's Bazar Student Forum, CUET.
- **Coordinator** – Cox's Bazar Student Society.
- **Senior Mathematics & Physics Instructor** – Tutoring.
- Organized tech seminars, student welfare initiatives, and academic mentoring.